WIA3002 | ACADEMIC PROJECT 1

FINAL REPORT

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TITLE:

SMARTPHONE BASED PALM OIL RIPENESS DETECTION USING LIGHTWEIGHT VISION MODELS

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ABSTRACT

As one of the world's top producers of palm oil, Malaysia's agricultural economy relies heavily on the quality of oil palm Fresh Fruit Bunches (FFB) processed at palm oil mills. The accurate determination of FFB ripeness at the factory loading ramp is a critical factor in maximizing the mill's Oil Extraction Rate (OER) and calculating fair payments to suppliers. Currently, ripeness assessment during FFB reception relies on manual visual inspection by Quality Control (QC) graders. This process is subjective, labor-intensive, and prone to human error, frequently leading to grading disputes between mill operators and suppliers.

This project presents a smartphone-based computer vision system designed to automate FFB ripeness classification at factory settings using lightweight deep learning models. By leveraging a MobileNetV2 architecture optimized via TensorFlow Lite (TFLite), the model categorizes FFB into four distinct MPOB-aligned ripeness stages. The methodology includes a two-stage transfer learning process, severe data augmentation, and FP16 quantization to compress the model for offline edge deployment on portable devices used by QC graders.

The final system achieved a Top-1 accuracy of 68.14% and a highly reliable Top-2 accuracy of 89.06%, proving the model understands the continuous spectrum of fruit ripeness. Maintained to perform under simulated ±30% lighting variations typical of semi-outdoor factory loading bays, the 4.26 MB model is capable of real-time offline inference (~13.87 ms per frame), providing mill graders with a robust, instantaneous decision-support tool to ensure high-quality processing.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Palm oil is one of Malaysia's most important agricultural commodities and plays a major role in the national economy. Malaysia, together with Indonesia, is among the largest producers and exporters of palm oil globally. The Malaysian Palm Oil Board (MPOB) dictates strict grading standards for FFBs at the reception point (loading ramp) of these mills to ensure a high and competitive Oil Extraction Rate (OER).

When lorries deliver FFBs to the factory, mill graders must sample and grade the bunches. If a batch contains too many unripe or underripe bunches, the oil content is drastically lower. Conversely, overripe bunches increase the free fatty acid (FFA) content, degrading the oil's market value. Therefore, accurate ripeness classification is critical in maximizing yield and maintaining product standards.

Currently, ripeness assessment is performed manually by plantation workers using visual inspection methods such as fruit color observation and loose fruit detection. However, this process is subjective, inconsistent, and highly dependent on worker experience and environmental conditions. These limitations often result in variation in harvesting decisions across different workers and plantations.

Recent advances in artificial intelligence (AI) and computer vision have enabled automated image-based classification systems that can assist in agricultural decision-making. In particular, lightweight deep learning models allow deployment on mobile devices, enabling real-time inference directly in the field without reliance on cloud computing.

1.2 Problem Statement

The quality of Fresh Fruit Bunches (FFBs) that are supplied directly to the palm oil mills (POM) affects the oil extraction rate (OER) and production efficiency. The grading process should be objective, consistent and transparent at the loading ramp to provide an accurate assessment of fruit ripeness and allow for fair compensation for suppliers while ensuring efficient mill operations. But the existing grading system is still largely based on manual visual inspection done by Quality Control (QC) grader.

Ripeness assessment by hand is a subjective and human error-prone process. Environmental conditions, fatigue and grader experience can impact the consistency of the grading operation. Furthermore, loading ramps are often situated in semi-outdoor areas where light levels can vary with changing sunlight levels, shadows cast by vehicles and reflections from nearby buildings. These conditions can make it difficult to accurately grade the product visually, and can result in different grades. This can lead to conflicts between millers and suppliers over fruit quality and related price deductions.

The automated classification of fruit ripeness has been shown to be a promising application of recent developments in deep learning and computer vision. But many of the available options

are based on models that are hard to compute, need high-performance hardware or cloud processing. These requirements make them impractical for use in a palm oil mill since they require a portable and responsive device that can be used directly in the field.

This project aims to overcome these problems by developing a smartphone-based palm oil FFB classification system using a lightweight deep learning model. The model will be optimized for on-device inference using TensorFlow Lite so that it can be used to classify the ripeness of a fruit in real-time and offline on an Android smartphone. The proposed system will be able to offer objective and consistent grading support, which will help to minimize the variability of grading and eliminate the dispute between stakeholders, and enhance the efficiency of the FFB reception process at palm oil mills.

1.3 Research Objectives

To address the industrial and technological gaps identified, this project outlines the following core research objectives:

1. To develop a smartphone-based oil palm Fresh Fruit Bunch (FFB) ripeness detection system using lightweight image classification models
2. To analyze and compare the performance of different lightweight deep learning architectures (such as MobileNet, EfficientNet-Lite, and Vision Transformers) for multi-class ripeness classification.
3. To evaluate the proposed models under varying lighting conditions typical of mill loading bays and optimize the best-performing model for completely offline smartphone deployment.

1.4 Scope of the Project

The study is aimed at visual classification of oil palm Fresh Fruit Bunches (FFB) into four ripeness classes which are adopted by the industry for grading purposes, namely Unripe, Underripe, Ripe and Overripe. The classification process is only performed with RGB images taken by smartphone cameras.

The proposed system is targeting the Android platform since it is widely used and easily available among plantation workers and field operators. The study takes an approach of Edge AI, where all inference processes are done on the mobile device itself for practical deployment in plantation environments. As a result, the application does not use cloud servers, an internet connection, or external application programming interfaces (APIs) for classification.

The scope of Final Year Project 1 (FYP 1) includes collecting and preparing the data for the project, preprocessing the images, augmenting the images, building a baseline classification model using transfer learning with the MobileNetV2 network, evaluating the model performance under simulated lighting changes, and converting the trained model into TensorFlow Lite (TFLite) format for deployment on mobile devices.

1.5 Project Schedule

Week	Planned Activities
Week 1	● Briefing for Academic Project 1 ● Preliminary exploration of potential project topics
Week 2	● Discussion with supervisor about FYP Title ● Confirmed FYP Title with supervisor
Week 3	● Conduct literature review focusing on project background and motivation
Week 4	● Continue literature review (related search papers/previous works) focusing on architecture ● Define problem statement and objectives
Week 5	● Start finding potential collaborators and project stakeholders ● Identify initial functional and non-functional requirements
Week 6	● First meeting with collaborators to align on project scope ● Gather and analyze initial information and context from the collaborators ● Gather initial dataset of palm oil images
Week 7	● Begin drafting Report 1 (Introduction & Literature Review Chapters) ● Perform data preprocessing and image augmentation setup
Week 8	● Draft methodology section for Report 1 ● Consolidate project planning for monitoring session ● Set up deep learning environment and baseline model architecture (MobileNetV2 feature extraction)
Week 9	● Submit Final Year Project Monitoring Slides (Monitoring Session)
Week 10	● Refine project based on monitoring feedback ● Fine-tune base model and implement Label Smoothing for robustness to lighting variations
Week 11	● Experiment with model optimization ● Evaluate model accuracy and confusion matrix ● Export preliminary model to TFLite (FP32/FP16) to demonstrate mobile feasibility
Week 12	● Finalize Final Report 1 writing (compiling Introduction, Literature Review, Methodology and System Analysis & Design) ● Prepare Slides for Viva presentation
Week 13	● AP1 Viva Presentation ● Final revisions to Final Report 1 based on viva feedback ● Submit Final Report via Ilmiah System

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The application of deep learning in agriculture has seen significant growth. A review of recent literature highlights various attempts to automate palm oil and general fruit ripeness classification, revealing both technological advancements and persistent limitations regarding edge deployment and real-world robustness.

2.2 Comparison of Existing Studies

To provide a clear overview of the current technological landscape, Table 2.1 summarizes the recent studies, highlighting their methods, results and critical limitations.

Table 2.1: *Comparison of recent studies on palm oil and fruit ripeness classification*

Reference	Dataset	Method / Algorithm	Results	Limitations / Research Gap
Suharjito et al. (2021)	400 field images (augmented to 4,000+)	MobileNetV1 and EfficientNet-B0 (via TensorFlow Lite)	EfficientNet-B0 achieved **89.3%** accuracy with 96 ms inference time.	Relied heavily on synthetic data augmentation; raises concerns about model robustness in actual plantation environments.
Soetrisno et al. (2023)	800 images (4 categories)	EfficientDet architectures (Lite0, Lite1, Lite2) via AutoML	EfficientDet-Lite2 achieved best mean Average Precision (mAP) of **0.84**.	The model struggled to distinguish between ripe and overripe fruit bunches due to highly similar visual characteristics.
Chong et al. (2025)	1,324 web images + 30 field images	MobileNet trained via transfer learning (Google	Reached **98%** overall accuracy across most categories.	Heavy reliance on web images with restricted viewing angles reduces confidence in real-world generalization.

		Teachable Machine)		
Setiawan et al. (2025)	400 images expanded to 1,000 (mobile + internet)	Custom Convolutional Neural Network (CNN) architecture	Achieved **88.5%** test accuracy after optimization.	Exhibited noticeable overfitting issues, indicating limited adaptability to unseen environmental lighting conditions.
Espinoza et al. (2025)	Moderate generic fruit dataset	Compared EfficientNet-B0 against Vision Transformer (ViT-B/16)	EfficientNet-B0 achieved **94%**, outperforming the ViT at 92%.	Vision Transformers require significantly larger datasets and higher computational resources, making them unsuitable for mobile edge deployment.

2.3 Review of Existing Products

In modern Malaysian palm oil mills, there has been a push towards automated grading. Some massive commercial systems employ heavy, stationary camera rigs mounted above conveyor belts to scan FFBs post-loading. However, these systems are highly expensive, stationary, and prone to bottlenecks. There is a distinct lack of portable, smartphone-based software products designed to empower individual QC graders moving dynamically around the vast, chaotic FFB piles at the mill reception ramp.

2.4 Review of Applicable Deep Learning Algorithms

2.4.1 MobileNetV2

Deep Convolutional Neural Networks (CNNs) like ResNet and VGG16 are notorious for their high computational cost, requiring millions of parameters and high-end GPUs. Since this project targets edge deployment on factory graders' smartphones, such heavy architectures are unfeasible. Therefore, **MobileNetV2** was selected as the base architecture for this study.

MobileNetV2, introduced by Google researchers (Sandler et al., 2018), is an architecture specifically tailored for mobile and resource-constrained environments. It achieves high accuracy with a fraction of the computational power by utilizing two main architectural innovations:

1. **Depthwise Separable Convolutions:**
   Traditional convolutions perform spatial filtering and feature combining in a single step, which is computationally expensive. MobileNetV2 splits this process into two separate layers: a depthwise convolution (which applies a single filter to each input channel) and a pointwise convolution (a 1x1 convolution that combines the outputs of the depthwise layer). This factorization drastically reduces the number of mathematical multiplications and the overall model size by nearly 8 to 9 times compared to standard convolutions.

2. **Inverted Residuals and Linear Bottlenecks:**
   Traditional residual networks (like ResNet) connect layers with high numbers of channels, compressing them in the middle. MobileNetV2 reverses this design. It uses inverted residuals, expanding a low-dimensional input to a higher dimension to extract rich features, filtering it, and then projecting it back to a low-dimensional space using a linear bottleneck layer. The removal of non-linear activation functions (like ReLU) in the bottleneck prevents the destruction of information in smaller feature spaces. Furthermore, MobileNetV2 utilizes the ReLU6 activation function, which is robust for low-precision calculation, making it highly suitable for mobile quantization.

Previous work by Al-Khowarizmi et al. (2025) showed that MobileNetV2 with transfer learning achieves 94.5% accuracy for binary ripeness classification, but multi-class performance remains lower. Furthermore, Izzatina et al. (2025) demonstrated that using MobileNetV2 as a student in a knowledge distillation framework can boost multi-class accuracy to over 95% while keeping the model lightweight which is a promising direction for the project.

2.4.2 EfficientNet

While MobileNetV2 focuses on reducing parameters through depthwise convolutions, EfficientNet (introduced by Tan & Le, 2019) tackles the problem of how to scale a Convolutional Neural Network (CNN) to achieve better accuracy without unnecessarily exploding the computational cost.

Historically, researchers scaled CNNs by arbitrarily increasing one of three dimensions:

1. **Depth:** Adding more layers.
2. **Width:** Adding more channels/filters per layer.
3. **Resolution:** Feeding larger input images.

EfficientNet revolutionized this by introducing Compound Scaling. It uses a mathematical formula (a compound coefficient) to scale network width, depth, and resolution in a principled, balanced manner. The baseline network, EfficientNet-B0, uses Mobile Inverted Bottleneck Convolutions (MBConv) similar to the foundational architecture of MobileNetV2 combined with Squeeze-and-Excitation optimization.

For edge AI deployment in factory settings, variants like EfficientNet-Lite strip away complex operations (like Swish activations) to optimize execution on mobile CPUs. Including EfficientNet in this study provides a highly optimized baseline to compare against MobileNetV2, determining

if compound scaling yields better Top-1 accuracy for borderline ripeness detection without sacrificing the sub-second latency required at the loading ramp.

Previous study Suharjito et al. (2021) reported 89.3% accuracy with EfficientNet-B0 on mobile, but at 96 ms inference – slower than our MobileNetV2 baseline (~13.87 ms). Soetrisno et al. (2024) used EfficientDet-Lite2 (which uses an EfficientNet backbone) and achieved 84% accuracy with better feature fusion via Bi-FPN, but still struggled with ripe/overripe ambiguity.

2.4.3 Vision Transformers (ViT)

For over a decade, CNNs have dominated computer vision. However, the introduction of the Vision Transformer (ViT) by Dosovitskiy et al. (2020) represented a massive paradigm shift. Instead of using convolutional filters to slide over an image and extract local features, ViTs adapt the architecture used in Natural Language Processing (like ChatGPT) for images.

A Vision Transformer works by dividing an image into a grid of fixed-size patches (e.g., 16x16 pixels). It mathematically flattens these patches and treats them exactly like "words" in a sentence. It then utilizes a Multi-Head Self-Attention mechanism. This allows the model to look at the entire image at once and determine how each patch relates to every other patch, capturing global context incredibly well (e.g., linking the color of a palm fruit on the left side of the bunch to the texture of an empty socket on the right side).

The Trade-off: While ViTs often achieve higher accuracy than CNNs, they lack "inductive bias" (the built-in assumption that pixels close to each other are related, which CNNs possess). Therefore, Transformers generally require massively larger datasets to train effectively and are heavily computationally expensive. Including Vision Transformers in this project's future comparison serves as a critical stress-test: it will scientifically determine whether the superior global context of self-attention justifies the massive computational overhead, or if lightweight CNNs remain the only viable choice for budget factory smartphones.

2.5 Summary and Research Gap

The literature reviewed shows that high accuracy has been reported for classification of fruits and palm oil, but most of the existing models have limitations in real world applications. These are high computation cost, fixed camera system and low robustness in varying environmental conditions.

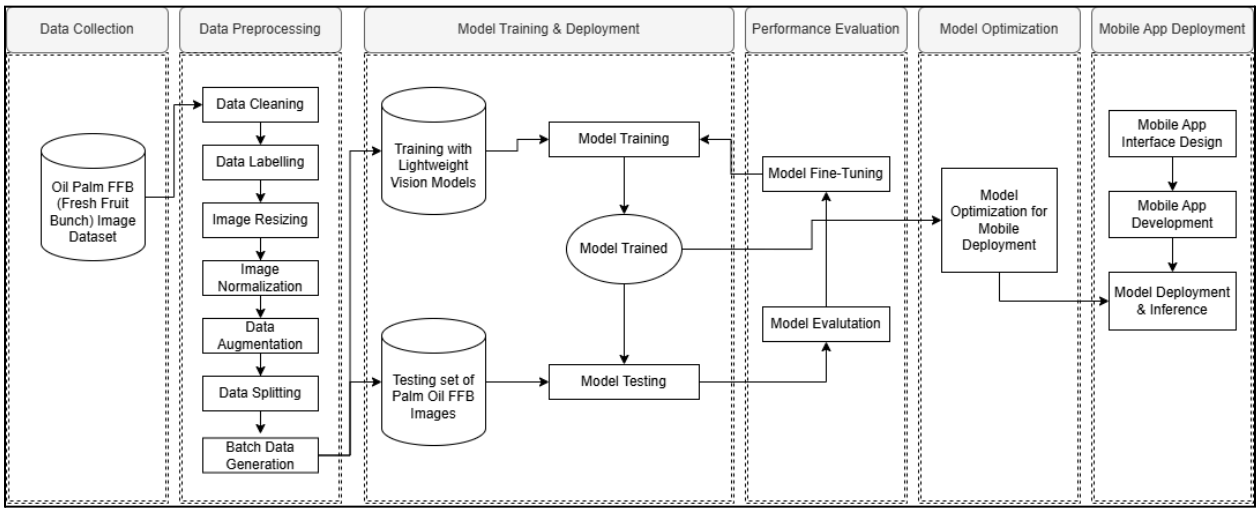
Hence, there is a research gap in the development of lightweight, mobile-ready deep learning systems that are both computationally efficient and robust to real world plantation environments. To fill this gap, this study introduces an edge AI solution using lightweight vision models (MobileNetV2, EfficientNet) for deployment on smartphones that can work under various lighting conditions with practical inference speed.

CHAPTER 3: METHODOLOGY

3.1 Overview of the Methodology

To achieve the research objectives, the project follows a structured System Development Life Cycle (SDLC): (1) Dataset Collection, (2) Data Preprocessing, (3) Model Training, (4) Performance Evaluation, (5) Model Optimization, and (6) Mobile App Deployment.

For FYP1, the focus is on developing a baseline deep learning model using MobileNetV2 and evaluating its performance on Fresh Fruit Bunch (FFB) classification. FYP2 will extend this work through model optimization, comparative analysis, and Android deployment.



System Development Life Cycle (SDLC)

3.2 System Development Approach

The methodology is divided into two phases:

FYP 1 (Completed):

- Dataset collection
- Data preprocessing and augmentation
- Baseline MobileNetV2 model development
- Model training and evaluation
- TensorFlow Lite conversion (preliminary)

FYP 2 (Planned):

- Model optimization (focal loss, fine-tuning)
- Knowledge distillation
- Architecture comparison (e.g., EfficientNet, ResNet)
- Android mobile application deployment

3.3 Dataset Collection

The foundation of any robust computer vision model is its data. The dataset utilized comprises 4,132 high-resolution images of Oil Palm Fresh Fruit Bunches (FFB) captured in various environments. The dataset was rigorously categorized into four classes aligned with the Malaysian Palm Oil Board (MPOB) grading standards: Overripe, Ripe, Underripe, and Unripe.

An exploratory data analysis (EDA) revealed a natural class imbalance. In real-world mill reception scenarios, batches are often heavily skewed towards unripe and underripe fruits due to premature harvesting practices by suppliers aiming to capitalize on high commodity prices. To ensure the model training was statistically sound, the dataset was split into three distinct subsets:

- **Training Set:** Used to update the model weights (approx. 70%).
- **Validation Set:** Used to tune hyperparameters and monitor overfitting during training (approx. 15%).
- **Testing Set:** A completely sequestered dataset of 521 images used exclusively for final, unbiased performance evaluation (approx. 15%).

3.4 Data Preprocessing and Augmentation

Raw images captured by varying smartphone cameras cannot be fed directly into a neural network. A strict preprocessing pipeline was established:

1. **Resolution Standardization:** All images were resized to a standard tensor dimension of 224 × 224 × 3 pixels, matching the input topology expected by the MobileNetV2 base model.
2. **Pixel Normalization:** Raw RGB pixel values ranging from [0, 255] were normalized to a range of [-1.0, 1.0]. This mathematical scaling prevents the vanishing/exploding gradient problem and allows the gradient descent optimizer to converge faster during training.

Advanced Data Augmentation:

To guarantee the model becomes invariant to the chaotic environments of factory loading bays, artificial noise and transformations were injected into the training set dynamically.

- **Spatial Transformations:** Random horizontal flipping and ±10% random rotations were applied. Since an FFB looks identical regardless of its orientation on the ramp, this forces the model to ignore spatial orientation.
- **Photometric Transformations:** Crucially, random brightness and contrast adjustments (varying up to 30%) were applied. Mill loading ramps are semi-outdoor environments affected by harsh afternoon sun glare and deep shadows cast by large lorry trailers. By manipulating the brightness during training, the model is forced to learn the physical, biological textures of the fruit rather than memorizing localized lighting anomalies.

Additional augmentation strategies that will be considered for future work:

1. **9-angle crop augmentation** (Suharjito et al., 2021) – rotating the camera angle by 0°, 45°, 90°, etc., to simulate different holding positions.
2. **Gamma correction** – instead of uniform brightness scaling, applying non-linear gamma values (0.4 to 1.6) to better mimic real shading from lorry canopies.

3.5 Model Development: MobileNetV2

MobileNetV2 is selected as the baseline model due to its lightweight architecture and suitability for mobile deployment.

It utilizes:

- Depthwise separable convolutions (reduces computation cost)
- Inverted residual blocks (efficient feature reuse)
- Pretrained ImageNet weights (transfer learning)

A custom classification head is added to adapt the model for the 4-class FFB classification task.

3.6 Model Training Strategy

Training a deep neural network from scratch requires massive datasets and excessive computational power. To bypass this, Transfer Learning was utilized. The MobileNetV2 model was pre-trained on the ImageNet dataset (containing over 1.4 million images), meaning it had already learned foundational visual features like edges, curves, and color gradients.

The training was executed in two distinct stages to protect these pre-learned features:

1. **Stage 1 (Feature Extraction - Frozen Base):**
   The entire pre-trained MobileNetV2 base was mathematically frozen (weights were locked). A custom classification head was appended to the top. This head consisted of a Global Average Pooling 2D layer, followed by a heavy 45% Dropout layer. Dropout randomly deactivates 45% of the neurons during each training step, forcing the network to distribute learning and aggressively preventing the memorization of training data (overfitting). Finally, a Dense layer with 4 nodes utilizing a Softmax activation function was used to output class probabilities.

2. **Stage 2 (Fine-Tuning):**
   Once the custom head converged, the top 30 convolutional layers of the MobileNetV2 base were unfrozen. The model was recompiled with a drastically reduced learning rate (1e-5). This delicate fine-tuning process allows the model to adjust its higher-order feature maps to specifically recognize palm oil textures (e.g., the spiky texture of unripe fruits versus the loose, detached sockets of overripe fruits).

Knowledge Distillation

To further improve classification performance while maintaining suitability for smartphone deployment, a Knowledge Distillation strategy will be investigated. In this approach, a larger and more powerful teacher model, such as ResNet50 or EfficientNet-B3, is first trained on the palm oil fruit dataset. The knowledge learned by the teacher model is subsequently transferred to a lightweight MobileNetV2 student model.

Unlike conventional knowledge distillation methods, this study will employ an ordinal distillation loss based on the work of Izzatina et al. (2025). The ordinal loss function incorporates the natural progression of palm oil fruit maturity (Unripe → Underripe → Ripe → Overripe), enabling the student model to better understand the relationships between adjacent ripeness classes. Previous studies have demonstrated that ordinal knowledge distillation can improve Top-1 classification accuracy by more than 10 percentage points while preserving the computational efficiency required for real-time mobile applications.

The performance of the baseline MobileNetV2 model and the distilled MobileNetV2 model will be compared to evaluate the effectiveness of the proposed knowledge distillation approach for smartphone-based palm oil fruit classification.

3.7 Handling Class Imbalance (Categorical Focal Crossentropy)

Class imbalance is a common challenge in agricultural image classification datasets, where certain ripeness categories may contain significantly more samples than others. When trained using a standard categorical cross-entropy loss function, deep learning models tend to become biased toward the majority classes, resulting in reduced performance on underrepresented classes.

To address this issue, this study proposes the use of **Categorical Focal Crossentropy** combined with class weighting during model training. Focal loss extends the conventional cross-entropy loss by introducing a modulating factor that reduces the contribution of easily classified samples while increasing the emphasis on difficult or misclassified examples. As a result, the model is encouraged to learn more discriminative features for minority classes, thereby improving classification performance across all ripeness categories.

In addition, class weights will be calculated based on the distribution of training samples and incorporated into the loss function to further mitigate class imbalance. This approach ensures that errors associated with underrepresented ripeness classes receive greater importance during optimization.

Previous studies on fruit ripeness classification have reported that focal loss with tuned parameters can improve recognition performance for minority classes. Therefore, focal loss parameters and class weights will be explored and optimized during the fine-tuning phase in FYP 2 to evaluate their effectiveness for smartphone-based palm oil fruit classification.

3.8 Model Optimization for Mobile Deployment

Deep learning models trained in TensorFlow utilize 32-bit floating-point (FP32) numbers to represent their internal weights and biases. While highly precise, FP32 requires significant memory storage and RAM, making it unsuitable for budget smartphones used in factory settings.

To solve this, Post-Training Quantization was applied. During the conversion from Keras (.h5) to TFLite (.tflite), all parameters were mathematically downcast to 16-bit floating-point (FP16) values. This halves the memory requirement of every single weight in the network. The result is a massive reduction in the model's footprint compressing the file size to just 4.26 MB while minimizing the loss of inference accuracy.

3.9 Offline Model Deployment Architecture

The ultimate goal of this project is to achieve accurate classification of Fresh Fruit Bunch (FFB) and practically deploy it in a real palm oil mill environment. Because the loading ramps are often located in areas with limited or unstable Internet connectivity, the proposed system is based on an idea of Edge AI, where all the image processing and classification will be done locally on the smartphone, without using cloud services.

In phase FYP 2, the chosen lightweight vision model will be converted to the TensorFlow Lite (.tflite) format and embedded in an Android app written in Kotlin. The suggested deployment procedure will go through the following four steps:

- The Android App will utilize the CameraX API to take pictures of Fresh Fruit Bunches straight from the smartphone camera.
- To make the consistency between training and deployment environment, images are captured and resized and normalized based on the local image preprocessing configuration set at the time of the model training.
- The preprocessed image tensor will be delivered to the embedded TensorFlow Lite Interpreter, which will do the classification using the computational resources of the smartphone (CPU, and if available, hardware accelerators like the NPUs).
- Result Presentation: The predicted ripeness category and confidence score will be presented in the mobile application interface, allowing for quick decision making by quality control staff.

The proposed solution enables the execution of the entire inference pipeline on-device, thereby mitigating the need for network connectivity, faster response time, and easier deployment in industrial settings. This design should enable real-time FFB ripeness assessment, while preserving the portability and accessibility benefits of smartphone-based systems.

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.1 System Overview

The proposed system is an end-to-end, smartphone-based image classification pipeline tailored specifically for palm oil factory settings. It is engineered to operate entirely offline (Edge AI), bypassing the need for complex IT infrastructure or Wi-Fi coverage across massive loading ramps.

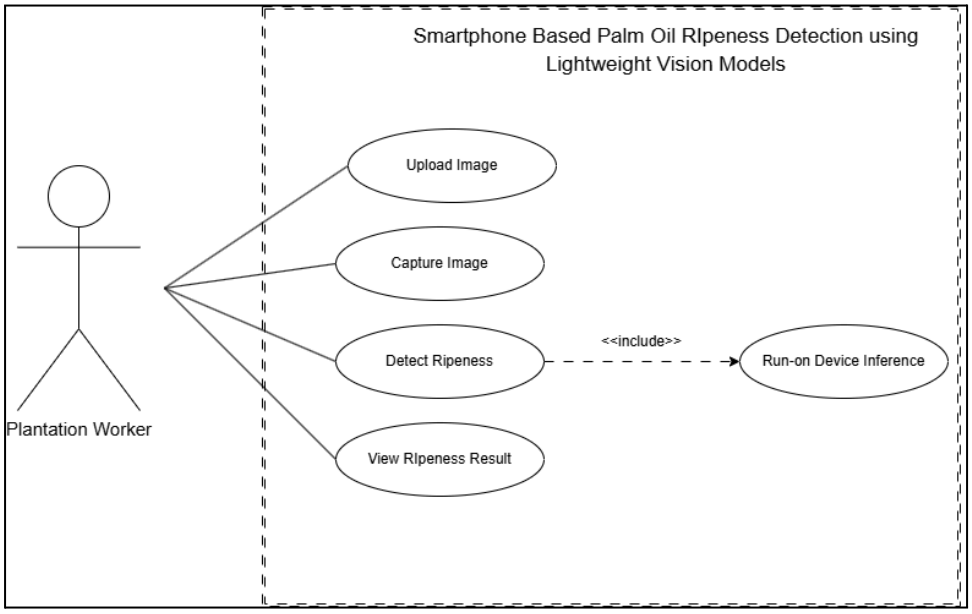
4.2 System Architecture and Modeling (UML)

The system workflow begins when the user (Mill Quality Control grader) either captures an image using the smartphone camera or uploads an existing image from storage. The image is then passed through a preprocessing module before being processed by the on-device inference engine. The trained model performs classification and returns the predicted ripeness category along with a confidence score.

A Use Case Diagram is developed to represent system interactions. The primary actor is the Mill QC Grader, who interacts with the system through the following use cases:

1. Capture Image
2. Upload Image
3. View Ripeness Result

Both the capture and upload use cases securely trigger the "<<include>> Run On-Device Inference" case, handled by the TFLite AI Engine



Use Case Diagram

4.3 Requirement Analysis

Systems requirements were defined based on operational needs in palm oil mill environments

Functional Requirements:

1. The system must allow graders to capture FFB images directly at the loading ramp.
2. The system must classify the image into 4 MPOB-standard ripeness categories (Unripe, Underripe, Ripe, Overripe).
3. The system must output the predicted class alongside a statistical confidence score to justify grading decisions to suppliers.

Non-Functional Requirements:

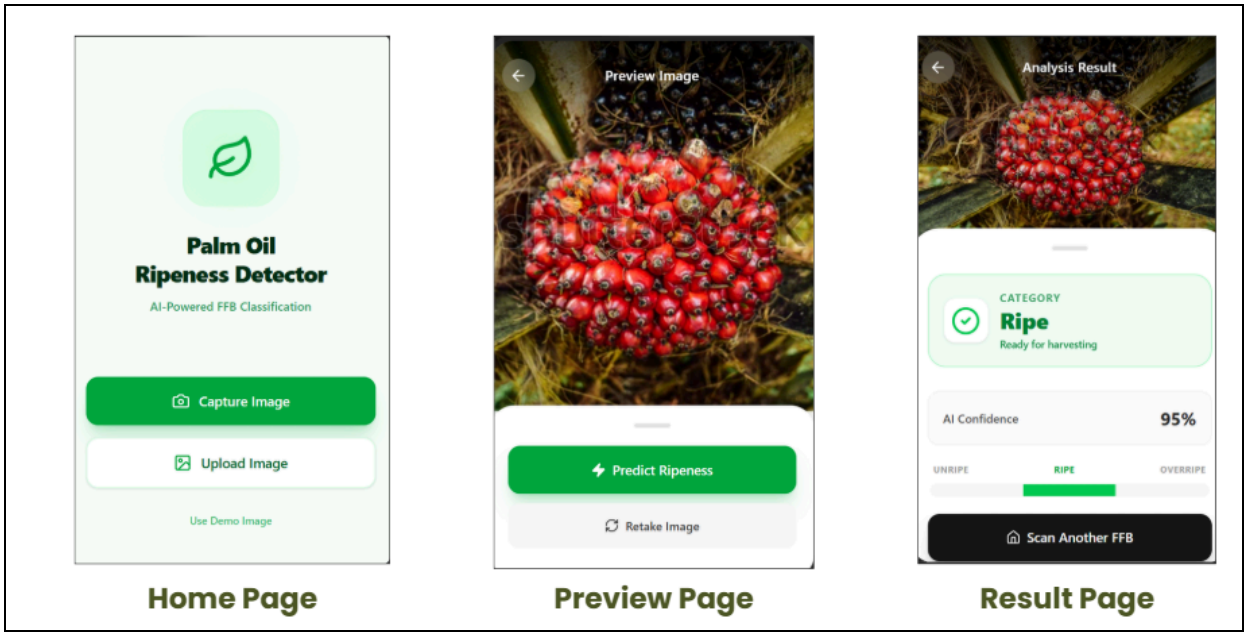
1. The model shall achieve a minimum target of ≥ 85% accuracy.
2. The system shall perform on-device inference within ≤ 2 seconds per image to maintain high factory throughput.
3. The optimized model size shall be ≤ 100 MB to ensure compatibility with mobile devices.
4. The system must maintain stable performance under varying semi-outdoor lighting conditions.

4.4 Prototype Application Interface Design

To ensure high usability for factory workers, the user interface (UI) mock-ups were designed to be simple and accessible:

- **Home Page:** Features prominent "Capture Image" buttons.
- **Preview Page:** Displays the selected image with a clear "Predict Ripeness" call-to-action.
- **Result Page:** Displays the output category in large, color-coded text (e.g., Green for Ripe, Red for Unripe) to provide immediate, undeniable visual feedback, accompanied by a confidence percentage bar.

The overall design prioritizes minimal interaction steps to accommodate real-time usage in factory conditions.



Mock-up Design

4.5 Development Tools

To ensure a robust and efficient development pipeline, a variety of industry-standard programming languages, frameworks, and Integrated Development Environments (IDEs) were utilized. The technical stack is divided into model development (FYP 1) and mobile deployment (planned for FYP 2):

1. Programming Languages:

- **Python:** Utilized as the core language for data preprocessing, data augmentation, and deep learning model training due to its extensive machine learning support.
- **Kotlin / Java:** The targeted languages for the native Android application development to ensure smooth hardware integration with the smartphone camera.

2. Frameworks and SDKs:

- **TensorFlow & Keras:** Used to build, train, and fine-tune the MobileNetV2 deep learning architecture.
- **TensorFlow Lite (TFLite):** Specifically used to perform FP16 Post-Training Quantization, converting the heavy Keras model into a compressed, mobile-friendly format.
- **Android SDK:** Essential for compiling the final mobile application for Android-based industrial tablets and smartphones.

3. Libraries:

- **OpenCV & NumPy:** Utilized for high-performance image processing, resizing, and matrix manipulations.
- **Matplotlib:** Used to generate evaluation graphs (such as the ROC curve and Training vs. Validation loss charts).

4. Integrated Development Environments (IDEs):

- **Google Colab:** Chosen for model training as it provides free access to cloud-based GPU acceleration, significantly reducing the time required to train the CNN over multiple epochs.
- **Visual Studio Code (VS Code):** Used for local script testing, debugging, and version control.
- **Android Studio:** The official IDE used for designing the mobile UI mock-ups and ultimately building the standalone APK file for factory field testing.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This research established a baseline smartphone-based ripeness classification system for oil palm Fresh Fruit Bunches (FFB) using the lightweight MobileNetV2 architecture. The study addressed the limitations of conventional manual grading practices by investigating the feasibility of an Edge AI solution capable of performing classification directly on mobile devices.

The developed model achieved a Top-1 accuracy of 68.14% and a Top-2 accuracy of 89.06% on an unseen testing dataset. While the Top-1 metric establishes an initial baseline, it is highly encouraging given the continuous biological nature of FFB ripening, which inherently causes high visual similarity between adjacent ripeness classes (e.g., late-stage underripe versus early-stage ripe). The model's validity is strongly reinforced by the robust Top-2 accuracy, demonstrating that the architecture successfully captures the correct ripeness trajectory and rarely makes severe misclassifications. Experimental evaluation further demonstrated robustness under simulated lighting variations commonly encountered at palm oil mill loading ramps. Through FP16 TensorFlow Lite quantization, the trained model was successfully compressed to 4.26 MB while maintaining fast inference performance suitable for smartphone deployment.

The findings of this FYP 1 phase demonstrate the technical feasibility of smartphone-based FFB ripeness assessment and provide a mathematically sound foundation for further optimization such as the implementation of focal lossin the next development phase. The results also support the potential of Edge AI technologies to improve grading consistency, reduce dependency on subjective visual inspection, and enhance operational efficiency within the palm oil industry.

5.2 Future Work

To transition this prototype into a fully deployable industrial tool, the following enhancements are planned for FYP 2:

1. **Architecture Comparison:** Evaluate and benchmark the current baseline against EfficientNet-Lite and Vision Transformers to maximize Top-1 accuracy.
2. **Mobile App Integration:** Embed the quantized .tflite model into a functional Android application.
3. **Knowledge Distillation** – Train a ResNet50 teacher and distill into MobileNetV2 using ordinal loss, aiming to boost Top-1 accuracy to >85% while keeping model size under 10 MB.
4. **Factory Field Testing:** Conduct live field testing and User Acceptance Testing (UAT) at a functioning palm oil mill loading ramp.

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